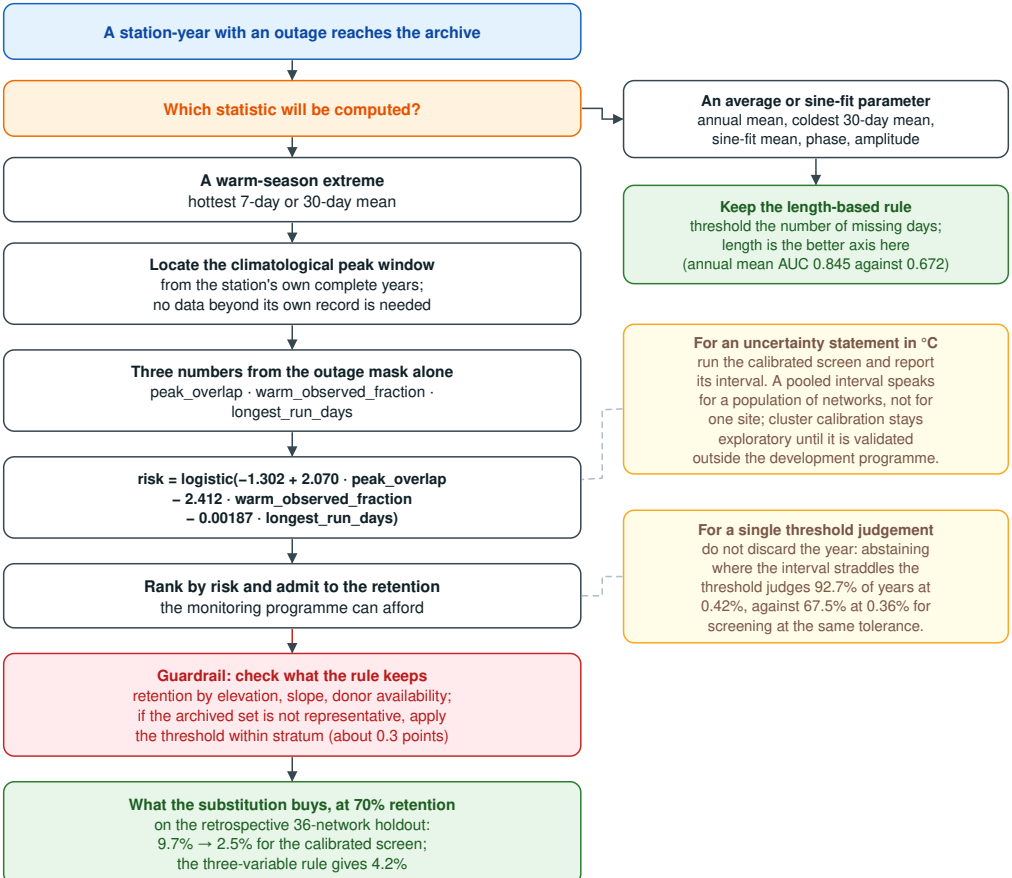
A station-year with an outage reaches the archive
Which statistic will be computed?
An average or sine-fit parameter
annual mean, coldest 30-day mean,
sine-fit mean, phase, amplitude
Keep the length-based rule
threshold the number of missing days;
length is the better axis here
(annual mean AUC 0.845 against 0.672)
A warm-season extreme
hottest 7-day or 30-day mean
Locate the climatological peak window
from the station's own complete years;
no data beyond its own record is needed
Three numbers from the outage mask alone
peak_overlap · warm_observed_fraction ·
longest_run_days
For an uncertainty statement in °C
run the calibrated screen and report
its interval. A pooled interval speaks
for a population of networks, not for
one site; cluster calibration stays
exploratory until it is validated
outside the development programme.
risk = logistic(−1.302 + 2.070 · peak_overlap
− 2.412 · warm_observed_fraction
− 0.00187 · longest_run_days)
For a single threshold judgement
do not discard the year: abstaining
where the interval straddles the
threshold judges 92.7% of years at
0.42%, against 67.5% at 0.36% for
screening at the same tolerance.
Rank by risk and admit to the retention
the monitoring programme can afford
Guardrail: check what the rule keeps
retention by elevation, slope, donor availability;
if the archived set is not representative, apply
the threshold within stratum (about 0.3 points)
What the substitution buys, at 70% retention
on the retrospective 36-network holdout:
9.7% → 2.5% for the calibrated screen;
the three-variable rule gives 4.2%